Lecture No 1

Pre-disaster preparedness is the practice of preparing for and planning to respond to a disaster. The goal of pre-disaster preparedness is to reduce the impact of disasters and support civil protection operations.

Some pre-disaster preparedness activities include:

- **Identifying disaster-prone areas**
- **Studying disaster-prone areas**
- **Collecting information about disaster intensity**
- **Developing early warning and evacuation systems**
- **Formulating emergency preparedness plans**
- **Conducting education and training**
- **Creating mass awareness about disaster management**
- **Updating egress plans**
- **Assembling a team to create a business continuity plan**

Pre-disaster preparedness can help communities build resilience, respond to disasters, and recover more effectively. It can also help communities access state and federal disaster resources more quickly

Managing a disaster is about mitigating, or avoiding, the risks of loss, providing prompt and adequate assistance to those affected, and ensuring a rapid and effective recovery. The concept of the Disaster Management Cycle integrates isolated attempts on the part of different government and nongovernment actors, towards vulnerability reduction or disaster mitigation, within the enveloping domain of disaster management. Governments, businesses, and civil society engage in the Disaster Management Cycle as a means of planning for and mitigating disasters, responding to disasters during and after they occur, and reclaiming the loss following a disaster. By acting appropriately at all points in the cycle, we are better able to prepare, give better warnings, reduce vulnerability, or prevent disasters during the subsequent iteration of the cycle. The Disaster Management Cycle can be divided into three stages : the pre-disaster stage, the during-disaster stage, and the postdisaster stage.

Pre-Disaster Stage:

- Preparedness, Prevention and Mitigation are the major activities in the pre-disaster stage. Prevention is always better than cure, so this program adheres to the principle. In this stage, various preventive measures and activities are undertaken well in advance so as to respond to disasters in an effective way.
- If the early warnings can be seriously heeded, much of the disaster could have been be avoided. The focus should therefore be on preparing communities and buildings to

withstand disasters. There are events of cyclones in India quite frequently, and warnings are generally given before the event occurs. If operative activities can be undertaken well in advance, then it becomes easy to prevent huge loss in terms of lives and property, the aftermath phase.

During-disaster Stage:

- Response and Relief are the important activities in the during-disaster stage. It starts in the aftermath of a disaster.
- The immediate aspects of an aid response include the search, rescue, and evacuation, identification of dead bodies, the disposal of debris, providing first aid, food, water, shelter, safety and security, health care, and sanitation, and restoring basic facilities.

Post-disaster Stage:

- The major activities in the postdisaster phase include: Rehabilitation, Reconstruction and Recovery. Disasteraffected communities will be made resilient and able to return to normalcy through these activities**.**
- Generally, this phase takes a long time as the efforts are made to restore all essential facilities to pre-disaster status.
- The major focus of this phase is on measures that could pave the way for the long-term recovery of social, economic, and physical structures, as well as processes in such a way that future disasters are unable to impact severely and irreversibly.

Disaster preparedness programmes aim to improve the technical and management capabilities of governments, organisations, and communities in order to reach a suitable degree of readiness to respond to any emergency crisis. These precautions, which are described as being logistically prepared to handle disasters, can be strengthened by having reaction mechanisms and processes, practising them, creating long- and short-term strategies, educating the public, and constructing early warning systems.

The number of natural disasters has increased fivefold over the last 50 years. Whether it's a storm, fire, flood, tornado, earthquake, or hurricane, your organization is more at risk of being damaged by a climate-related event now than ever before.
Think your particular organization won't be affected? Well, between January 2013 and January 2023, 88.5% of all U.S. counties declared a natural disaster. And a disaster can cost you big bucks.
Economic losses from storms, floods, fires, etc. have increased sevenfold from the 1970s to the 2010s. From 2010-2019, natural disasters caused $383 million of losses a day on average. In the 70s, losses were only $49 million per day on average.
Scared yet? You should be. Natural disasters can damage your facilities, hurt your employees and clients, stop your productivity in its tracks, and even put you out of business entirely.
Forty percent of businesses do not reopen after a natural disaster. Another 25% of businesses fail within a year after a disaster and an additional 90% fail within two years.

However, there is a bright spot in all this doom and gloom. Organizations who plan ahead for natural disasters can react and begin recovering more quickly.
But as a facilities manager or building owner, how do you plan for something as catastrophic as a fire, flood, or hurricane? Where do you even begin?
That's where the four phases of disaster management come in. They're a useful framework to build your disaster recovery and business continuity plans around. An up-to-date facility condition assessment is another important tool for disaster management that, as you'll read below, plays a key role in all four phases.

The 4 Phases of Disaster Management

Think of disasters as recurring events that take place in four main phases:

1. Mitigation

2. Preparedness

3. Response

4. Recovery

All organizations are in at least one phase at any given time. Understanding these four phases empowers your organization to prepare for and respond to crises in a smarter, more informed way. Making the right decisions gives your organization the best chance at survival and recovery following a climate-related event.
Let's take a closer look at what the disaster management phases mean and how a facility condition assessment helps you in each one.

Phase 1: Mitigation

Meaning: To prevent future emergencies and take steps to minimize their effects
The mitigation phase occurs **before** a disaster takes place. Here, an organization will take steps to protect people and property, while also decreasing risks and consequences from a given disaster situation. The organization's main goal is to reduce vulnerability to disaster impacts (such as property damage, injuries, and loss of life).
Examples of mitigation may can include measures such as:

- Clearing space around buildings to create a defensible space against fires
- Adding levees or improving property drainage to protect from flooding
- Securing furniture to floors and walls to help prevent damage/injuries during earthquakes
- Re-locating structures to less disaster-prone areas

Phase 2: Preparedness

Meaning: To take actions ahead of time to be ready for an emergency
The **preparedness** phase also occurs **before** a disaster takes place. Here, an organization attempts to understand how a disaster might affect overall productivity and the bottom line. The organization will also provide appropriate education while putting preparedness measures into place.

Examples of preparedness include hosting training, education, drills, tabletop exercises, and full-scale exercises on disaster preparedness. This ensures that stakeholders know what to do in the event of an emergency. Updating egress plans also falls under preparedness.
Organizations may also assemble a team to create a business continuity plan and list of resources needed to recover from a disaster.

Phase 3: Response

Meaning: To protect people and property in the wake of an emergency, disaster, or crisis
The **response** phase occurs in the **immediate aftermath** of a disaster. Organizations must focus their attention on addressing immediate threats to people, property, and business. Occupant safety and wellbeing largely depends on your preparedness levels before disaster strikes.
The most notable example of the response phase is ensuring that people are out of harm's way. The organization can then move on to assess damage, implement disaster response plans, triage cleanup efforts, and start resource distribution as necessary.
Businesses will also need to navigate building closures, preliminary damage assessments, and hampered communication with stakeholders (like staff, vendors, and suppliers) due to shutdowns.
As the response period progresses, focus will typically shift from immediate emergency issues to conducting repairs, restoring utilities, re-establishing operations, and cleaning up. The organization will also need to begin planning the reconstruction of damaged infrastructure.

Phase 4: Recovery

Meaning: To rebuild after a disaster in an effort to return operations back to normal
The **recovery** phase takes place **after** a disaster. This phase is the restoration of an organization following any impacts from a disaster. By this time, the organization has achieved at least some degree of physical, environmental, economic and social stability.
The recovery phase of a disaster can last anywhere from **six months to a year** (or even longer depending on the severity of the incident).
An example of recovery is creating strategic protocols and action plans to address the most serious impacts of a disaster. The protocols should give clear steps to follow for various disaster events and cover multiple scenarios.
For example, a flood protocol could include:

- The organization's internal plan to mitigate the effects of water (fan deployment, etc.)
- The preferred water remediation service and their contact information
- If that vendor is unavailable, secondary and tertiary services to contact

In this phase, the organization works to obtain new resources, rebuild or create partnerships, and implement effective recovery strategies. The organization also takes steps to reduce financial burdens, rebuild damaged structures, and reduce vulnerability to future disasters.

Lecture No 2

Hazard zonation maps are created by dividing an area into zones based on the likelihood of a hazard occurring. The process involves analyzing the physical characteristics of an area and considering environmental and triggering factors.

Here are some steps for preparing hazard zonation maps:

- **Identify factors**: Consider environmental factors like slope angle, soil type, and geological structures, as well as triggering factors like rainfall intensity and earthquakes.
- **Analyze data**: Use satellite and other data to map the factors.
- **Assign weights**: Use software to translate the factors into weights.
- **Create zones**: Use the weights to create zones with varying degrees of hazard.
- **Overlay maps**: Use GIS to overlay the derived maps to create the hazard zonation map.
- **Clarify in the field**: Make adjustments based on field clarifications.

Hazard zonation maps can help with disaster mitigation by: Identifying high-risk areas, Developing early warning systems, Implementing mitigation measures, and Developing land-use policies and regulations

Introduction:

Disaster preparedness refers to measures taken in advance to minimize the impact of natural or man-made disasters on human life and property. Disaster preparedness is crucial for effective disaster management. Hazard zonation mapping is an important tool that can aid in disaster mitigation, especially in the case of landslides.

Hazard zonation mapping can help in disaster mitigation in the case of landslides in the following ways:

- **Identification of High-Risk Zones:** Hazard zonation mapping can help identify areas that are susceptible to landslides.
- By analyzing the physical characteristics of an area, such as slope angle, geology, soil type, rainfall, and vegetation cover, different zones can be identified with varying degrees of susceptibility to landslides. This information can be used to prioritize disaster mitigation efforts in high-risk zones.
- **Development of Early Warning Systems:** In areas identified as high-risk zones, early warning systems can be developed to alert residents to evacuate in case of a landslide.
- By using hazard zonation mapping, authorities can identify critical areas where early warning systems can be installed, such as near rivers or steep slopes.
- **Implementation of Mitigation Measures:** Hazard zonation mapping can help disaster managers to take proactive measures to mitigate the impact of landslides.

- In high-risk zones, measures such as constructing retaining walls, stabilizing slopes, and creating drainage systems can be implemented to prevent or minimize the risk of landslides.
- **Development of Land-Use Policies and Regulations:** By identifying high-risk zones, hazard zonation mapping can help in the development of land-use policies and regulations.
- Restrictions can be put in place to limit the use of high-risk areas for certain activities, such as building construction, mining, or logging. Such policies can help to prevent or minimize the risk of landslides and other disasters in vulnerable areas.

Conclusion:

Hazard zonation mapping can help in disaster mitigation by identifying high-risk areas for landslides and allowing for appropriate measures to be put in place to reduce their impact. By prioritizing disaster preparedness and taking advantage of such tools, we can better protect our communities and minimize the loss of life and property in the face of natural disasters.

Forecasting and warning are crucial components of disaster management, and early warning systems (EWS) are used to integrate these elements:

Component	Description
Forecasting	Involves making a range of possible outcomes for the future, based on past data
Warning	Involves communicating the forecast to the public, usually through credible sources like government agencies or media
Early warning system	An integrated system that combines forecasting, warning, and other elements to help people take action before a disaster

EWSs use sensors to monitor potential hazards, and issue warnings when certain measurements are taken. This gives communities time to prepare and implement disaster management plans. EWSs can help reduce the damage caused by disasters by up to 30% if they're issued within 24 hours.

EWSs can be used for a variety of hazards, including: Floods and droughts, Water-related hazards, Earthquakes, Geohazards, and Tsunamis and coastal hazards.

Introduction The terms prediction and forecast are synonyms, often used interchangeably. The word prediction is of Latin origin, a hybrid of words prae (meaning, before) and dicere (meaning, to say). Prediction, therefore, literally means saying something beforehand. The term forecast is also made of hybrid of words fore (meaning, front) and cast (meaning, to throw). The term forecast therefore means to throw something upfront.

When we talk of prediction and forecasting in the context of natural hazards, for example earthquakes, The difference is that in prediction and forecasting is that in prediction we make a binary statement (one single outcome)—whether earthquake will occur, or it will not occur.

On the other hand, forecast will state a range of possible outcomes for the future in terms of probability of occurrence of an event, like probability of occurrence of an earthquake at a specific location within a certain specified magnitude range and in a particular window of strike time. Such forecasts are usually based on projection of the past into the future through in-depth study and analyses of the available data.

Predictability

• Predictability of a disaster is the key to understand its nature and thereby to assess the chances of its occurrence and the fury of the event.

• For man-made disasters, it is the human error or mechanical fault or organizational failure that is responsible. Therefore, there is no concept of predictability as such for man-made disasters.

• Mock drills, regular inspections and updating of precautionary measures take the place of predictability, forecasting and warning in case of man-made disasters.

Forecasting

• For natural disasters that have a fair amount of inherent predictability, forecasting is the next step in disaster management.

• Forecasting has to be based on sound scientific principles and operationally proven techniques.

• It has to be done by authorized agency or individual. In order to be effective, the • Forecast has to be clearly worded and it should be transmitted quickly to the user.

Warning

• Once a forecast is available regarding an anticipated disaster event, it has to be converted quickly into an area-specific and time-specific warning.

• The warnings also need to be user-specific because the capacity of different users to withstand 'the impacts of a disaster arc different.

• A warning has no value unless it reaches the users quickly and well in time, Therefore, quick communication is very important at the warning stage,

A disaster preparedness plan can help you and your family stay safe during a disaster. Here are some tips for creating a disaster preparedness plan:

- **Identify risks**

  Consider the types of hazards that could affect your area, and what impact they might have.

- **Create a plan**

  Work with your family to develop a plan that includes basic steps for addressing potential hazards. Keep the plan simple so it's easy to remember.

- **Practice**

  Practice your plan with your family, including evacuating your home.

- **Prepare a kit**

  Put together an emergency kit for your home and car that includes non-perishable food, water, a flashlight, batteries, first aid kit, and more.

- **Prepare your home**

  Eliminate hazards in your home and workplace.

- **Prepare your vehicle**

  Include a flashlight, extra batteries, maps, first aid kit, and more in your vehicle.

- **Prepare for the weather**

  If you live in a cold climate, prepare for the possibility of not having heat during or after a disaster.

- **Review and update**
  Review your plan every six months, and quiz your family on what to do. Restock food and water supplies every six months, and recharge your fire extinguisher.

Disaster Preparedness Tips

- Prepare to be self-sufficient for at least three days by putting together an emergency kit, including: non-perishable food, water, a flashlight, a portable, battery-operated radio or television, batteries, medicines, anti-bacterial hand wipes or gel, first aid kit, money, seasonal clothing, and sanitation supplies.
- Conduct practice drills so you and your family know the safe locations in your home for each type of emergency. Decide how and where your family will reunite if separated.
  Choose an out-of-state friend or relative that separated family members can call to report their whereabouts and conditions.
- Learn first aid and CPR from your local Red Cross chapter or other community organizations.
- Learn how to shut off gas, water and electricity in case the lines are damaged. Make sure insurance coverage is up-to-date and reflects present property values. Check on flood insurance.
- Compile an inventory of home contents. Take pictures and/or video. Store in a safe place.
- Check chimneys, roofs, walls and foundations for stability. Make sure your house is bolted to its foundation.
- Secure your water heater and major appliances, as well as tall, heavy furniture, hanging plants, picture frames and mirrors (especially those over beds).
- Make arrangements for pets.
- Organize your neighborhood to be self-sufficient after a disaster.

During a Disaster!

If you are evacuated:

- Follow directions of local officials. Carry your disaster supplies kit with you.
- Unplug appliances; turn off electricity, gas, and main water valve. (Safety note: do not attempt to re-light the gas pilot. Call the utility company.)
- If time permits, elevate or move furniture to upper floors.
- Tell someone outside of storm area where you are going.
- Lock home and leave.

If you stay at home:

- Listen constantly to a battery-powered radio or television.
- Stay inside away from windows, skylights and glass doors.
- If power is lost, turn off major appliances and keep refrigerators and freezers closed.

After a Disaster!

Unless there is an immediate life-threatening emergency, do not attempt to use the telephone. Be sure to:

- Stay calm. Check on neighbors, especially elderly or disabled.
- Turn on your portable radio or television for instructions and news reports. For you own safety, cooperate fully with public safety officials and instructions.
- Use a flashlight to cautiously check for gas and water leaks, broken electrical wiring or sewage lines. If there is damage, turn the utility off at the source. Immediately report gas leaks to your utility company. Check for downed power lines; warn others to stay away.
- Check your home for cracks and damage, including the roof, chimneys and foundation.
- Do not use your vehicle unless there is an emergency. Keep the streets clear for emergency vehicles. If you must drive, watch for downed power lines, flooded streets and highways and undermined roads.
- Take pictures of the damage, both house and contents, for insurance claims.
- When electricity is lost for several hours or days, frozen and refrigerated food may not be safe to eat. Do not re-freeze thawed food. Throw away all food that has been under flood waters, except canned food, but wash and sanitize the cans before opening. All food that cannot be saved should be double-bagged for normal trash disposal or buried at least 2-feet deep.
- Conserve water if your septic system is flooded.
- If your system lost pressure, boil water for 3 minutes before consuming.
- In warm weather, empty water out of birdbaths, tires, flower pots and other containers to limit mosquito larvae growth.

Things You Need:

1. Water - You need clean safe water to drink. Store 1 gallon per day for each person in your home for drinking and cooking. Experts say it is best to plan for three days. You can buy bottled water from the grocery store or bottle it yourself. If you bottle it yourself, choose a clean washed container like a soda bottle. Add four drops bleach (sodium hypochlorite) per quart of water. Don't use the scented bleaches. You may also

need water to flush the toilet if your home has a well with an electric pump. If you have any warning time before the power goes off, run your bathtub and your washing machine full of water for flushing the toilet. To flush the toilet when the power is off, just remove the tank lid and fill the tank with water. Now flush and the tank will empty as the water goes out of the bowel. Fill the tank back up for the next trip.

2. Food - You will need food that will not spoil or go bad if it is not kept cold. Have several days worth of canned foods on hand that could be eaten without heating if need be. Keep some foods that will keep for several days without heat or cold. Have on hand foods like peanut butter, crackers, fruit, vegetables, bread, and cereal. Food in your refrigerator and freezer will keep for a while depending on many factors. Keep the door closed as much as possible. Remember that you can cook on an outdoor grill if the weather is safe but you must do it outdoors.
3. Shelter - Staying dry and warm is most important in the case of winter storms. If your home has a safe fireplace, you have a good source of warmth for your family. Keep a stack of dry firewood in preparation for storm-related power failure. Gas logs will also work without power. Keep warm by dressing in layers. Wrap up in blankets for sleeping or sitting still. If you get wet, get dry as soon as possible. Make sure that your feet and hands stay warm and dry and check the hands and feet of children and older folks.
4. Transportation - In case of emergency, you need to be able to get to help, or get a message to help so that they can get you. If it is safe to drive your car, help is available at your community fire department. A Deputy Sheriff will be stationed at fire departments when communications lines are down. If you have to walk to help, remember to take care of yourself so you don't become the victim instead of the rescuer. Dress in layers, take water and food, and take the safest route even if it may take longer.
5. Radio - Most radio stations will broadcast even if your home power is off. Keep a battery powered radio available with extra batteries to receive emergency information.

Lecture No 3

Land use zoning is a tool that can help prevent and mitigate disasters by integrating natural hazard assessment into land use planning. Land use zoning can help reduce the risk of disasters by:

- **Identifying risk areas**

  Using risk assessment and mapping to identify areas with a high potential for natural hazards

- **Enforcing land use criteria**

  Using local regulations, building codes, and material specifications to enforce land use decisions

- **Monitoring and evaluating**

  Using disaster risk management indicators to monitor and evaluate the effectiveness of land use zoning

- **Involving stakeholders**

  Including relevant actors and organizational mechanisms in the decision-making process

Here are some other ways that land use planning can help with disaster management:

- **Planning for emergency measures**: Organizing emergency measures and coordinating recovery operations
- **Displacing populations**: Displacing populations if necessary
- **Reconstructing**: Reconstructing after a disaster

Land use planning can help reduce disaster losses in new development areas. It can also help strengthen local resilience by empowering vulnerable populations and designing socially acceptable solutions.

Zoning plays a key role in disaster management by dividing areas into zones based on the risk of disasters and the allowed land uses:

- **Zoning for flood risk**

  Zoning can help minimize the impact of flooding on people and infrastructure by dividing urban areas into zones with different development restrictions. For example, construction may be prohibited in high-risk areas like floodplains.

- **Zoning for seismic hazard**

  Zoning can help reduce the impact of earthquakes by designating areas with different densities based on geologic considerations. For example, areas with a high risk of ground failure from liquefaction may be designated as very low density residential.

- **Zoning for disaster prevention**

Zoning can help prevent disasters by relating natural hazard assessments to land-use planning. For example, micro-zoning can help determine land-use and building criteria by analyzing the probability of natural hazards in a given site.

Zoning can also help with: Organizing emergency measures, Coordinating recovery operations, Improving the capacity of the poor and most disadvantaged, and Promoting gender equality

Information, Education, and Communication (IEC) materials are used to educate people on how to stay safe and prepared during disasters. IEC materials can be used for a variety of disasters, including earthquakes, floods, and disaster risk reduction.

IEC materials are an essential part of any disaster management program. They help to create an alert community by promoting awareness and equipping people with the knowledge and skills they need to stay safe.

Some examples of IEC materials include: Be Red Cross Ready, Be Ready, Antiseismic Basic Guidelines, and Preparing for Disasters - Earthquakes.

The Center for Disaster Preparedness produces IEC materials that encourage people to consider the socio-economic, cultural, and political context of their communities when taking action to reduce and manage disaster risk.

Disaster Preparedness

Are the measures that ensure the organized mobilization of personnel, funds, equipment, and supplies within a safe environment for effective relief. Disaster preparedness is building up of capacities before a disaster situation prevails in order to reduce impacts. Its measures include availability of food reserve, emergency reserve fund, seed reserve, health facilities, warning systems, logistical infrastructure, relief manual, and shelves of projects. Disaster preparedness is just one element of a holistic approach to the reduction of risk associated with natural hazards. However, an adequate level of preparedness can be particularly essential to saving lives and livelihoods in the face of a natural hazard event.

Response

Response is the set of activities implemented after the impact of a disaster in order to assess the needs, reduce the suffering, limit the spread and the consequences of the disaster, open the way to rehabilitation. The provision of emergency services and public assistance during or immediately after a disaster in order to save lives, reduce health impacts, ensure public safety and meet the basic subsistence needs of the people affected.

Preparedness planning:

aims to establish a standing capacity to respond to a range of different situations that may affect a country or region by putting in place a broad set of preparedness measures. This includes for example early warning systems, ongoing risk and vulnerability assessment, capacity building, the creation and maintenance of stand-by capacities and the stockpiling of humanitarian supplies. Undertaking a contingency planning process will be a key component in developing an analysis of what needs to be done in this process, and will help in the designing, testing and implementation of response actions

Preparedness planning is a process that requires constant and sustained vigilance. Developing a preparedness capability is only the first step. Once the basic preparedness activities outlined in this plan have been undertaken, it is critical that human and other resources are allocated to

maintain and update systems, and that lessons learned in the wake of hazard events be used to strengthen preparedness in the future.

Early Warnings

The term 'early warning' is used in many fields to describe the provision of information on an emerging dangerous circumstance where that information can enable action in advance to reduce the risks involved. An Early Warning System (EWS) can be defined as a set of capacities needed to generate and disseminate timely and meaningful warning information of the possible extreme events or disasters (e.g. floods, drought, fire, earthquake and tsunamis) that threatens people ‘s lives.

Elements of Early warning

Early warning is the integration of four main elements:

1 Risk Knowledge: Risk assessment provides essential information to set priorities for mitigation and prevention strategies and designing early warning systems.

2. Monitoring and Predicting: Systems with monitoring and predicting Capabilities provide timely estimates of the potential risk faced by communities, economies and the environment.
3. Disseminating Information: Communication systems are needed for delivering warning messages to the potentially affected locations to alert local and regional governmental agencies. The messages need to be reliable, synthetic and simple to be understood by authorities and public.

4. Response: Coordination, good governance and appropriate action plans are a key point in effective early warning. Likewise, public awareness and education are critical aspects of disaster mitigation.

Need of Early Warning System

The first one, clearly, is public safety, and the protection of human lives. The second is the protection of the nation ‘s resource base and productive assets (infrastructure and private property or investments) to ensure long-term development and economic growth.

Prevention: is defined as those activities taken to prevent a natural phenomenon or potential hazard from having harmful effects on either people or economic assets. Delayed actions drain the economy and the resources for emergency response within a region. For developing nations, prevention is perhaps the most critical components in managing disasters, however, it is clearly one of the most difficult to promote. Prevention planning is based on two issues: hazard identification (identifying the actual threats facing a community) and vulnerability assessment (evaluating the risk and capacity of a community to handle the consequences of the disaster). Once these issues put in order of priority, emergency managers can determine the appropriate prevention strategies. Disaster prevention refers to measures taken to eliminate the root causes that make people vulnerable to disaster.

ROLE OF INFORMATION, EDUCATION, COMMUNICATION AND TRAINING

Information, Education and Communication (IEC) are of vital significance for disaster preparedness, prevention and mitigation. IEC are the essential constituents of any programme especially in a disaster related scenario.

Information

Preparing the community for disaster management through information implies the transfer of basic knowledge by means of data, practices and processes to the community so as to increase their awareness. It is useful for the administration and other agencies for planning purposes, and beneficial to the community for development of internal coping mechanisms. Information addresses the questions of What, When, Where, How and How Much with respect to the anticipated disaster.

Training

Training attempts to improve knowledge, skills and attitudes amongst the recipients. With regard to managing disasters, training is accorded a very important place, as it imparts the necessary knowledge and skills to varied categories of personnel. It improves the performance of people involved in managing the crisis. It enables them apply the knowledge and skills to situations. Training needs to be given to build, strengthen and enhance the capacities of individuals, organizations, and communities. It is necessary to have institutional support for effective training. Information Education Communication and Training activities are to be based on appropriate needs analyses. It needs to be holistic as there are several categories of personnel involved in disaster management activities. These include policy makers, government officials, specialists, technocrats, youth, NGOs, CBOs, communities etc. The training needs of these personnel differ. Accordingly, the IEC and training programmes have to be developed.

Education

Education implies learning that makes people aware and knowledgeable. Creativity addresses the values of the people in question and besides making them aware of the risks, 'it enlightens them about their issues, the actions needed to be taken and their roles, - and it also motivates them to act. Education is a continuous process throughout the human life that attempts to modify the attitudes, and behaviour of the people. Ed~~cation enlightens the people about the 'Why' aspect as to the occurrence of disasters. It tells them what to do and what not to do for preventing or mitigating disasters. It orients them about ways of managing situations, and provides them with several alternatives. People generally have a fatalistic attitude towards disasters, wherein disasters are considered as acts of god, and demonstrate helplessness to do anything to mitigate them. Education lessens this kind of attitude by making communities aware of causes of disasters, best available alternatives and means of mitigating them and enabling them to take counter disaster measures. The ways of imparting disaster education are (dependent on the target groups. Usually, it is the low-income group communities that an: most vulnerable to disasters and need to be educated. As such, the design and contents of the educational material have to be based on these considerations, and should be acceptable, understandable, interesting and motivating for the community. While educating, communities, their traditional practices, and wisdom, existing knowledge in coping with such situation-must also be kept in mind, and integrated with the educational process. Pit tempts need to be made to make the community understand the benefits of preparedness and mitigation, the relationship between the disasters and development. The aim of education is to promote an informed, alert and self-confident community that can cope with disasters,

Communication

Communication involves information exchange between two or more groups or individuals, It could be applied to both, education emergency communication management. In terms of disaster education and preparedness, communication is a process of knowledge transfer to the community regarding disaster- risks, preparedness and mitigation measures. In this process, the system is of flow of information from a singular awakened source to the mass group, i.e.

the community. Another manifestation of the same process is in times of emergency build-ups when the communication relates to hazards forecasts, at such times warnings may have to be issued to the community regarding the impending disaster in a clear understandable manner. This is a critical and final stage of pre disaster communication that gives the community a chance to take evasive action, or move away from the scene to a safer place in order to mitigate the disaster impact.

Once the disaster strikes, communication becomes an essential support system for relief as well as Search and Rescue (SAR) operations. Field teams have to be in contact with the base station or central cell. The efforts of various agencies need to be coordinated for a better and wider coverage in an efficient manner. An effective communication system facilitates various teams get to know what kind of assistance is required and in which areas. This is also useful for reaching out to the affected community with relief materials such as food, medicines and clothing. The field teams can also send back information to the base station regarding the extent and nature of additional assistance required, and the deficient areas. They should also inform about the areas covered, so that other teams may be notified not to visit the same areas again, thereby avoiding overlap. Special medical help or evacuation teams may be asked in critical situations, and valuable time could be saved in getting these if there is an on-line communication system, which would eliminate the need for physical communication of information. IEC attempts to create and strengthen the community awareness about which we shall discuss in the next section. Disaster communication management means the designing of appropriate communication strategy, procedures and contents in advance, according to the nature and objectives of the disaster, to smoothen its operations and avoid misunderstanding and other complicated problems which arise due to any communication gap. The purpose is to enlighten people likely to be affected by disaster so that timely preventive measures can be taken. Without the use of IEC activities disaster management cannot be effective.

Lecture No 4

The pre-disaster stage of disaster management is a phase of sustained action taken to reduce the risk of human life and property loss from natural disasters. The main goal of this stage is to reduce vulnerability to disaster impacts, such as injuries, property damage, and loss of life.

Introduction to Disaster Management Cycle

Managing a disaster is about mitigating, or avoiding, the risks of loss, providing prompt and adequate assistance to those affected, and ensuring a rapid and effective recovery. The concept of the Disaster Management Cycle integrates isolated attempts on the part of different government and nongovernment actors, towards vulnerability reduction or disaster mitigation, within the enveloping domain of disaster management. Governments, businesses, and civil society engage in the Disaster Management Cycle as a means of planning for and mitigating disasters, responding to disasters during and after they occur, and reclaiming the loss following a disaster. By acting appropriately at all points in the cycle, we are better able to prepare, give better warnings, reduce vulnerability, or prevent disasters during the subsequent iteration of the cycle. The Disaster Management Cycle can be divided into three stages : the pre-disaster stage, the during-disaster stage, and the postdisaster stage.

Pre-Disaster Stage:

- Preparedness, Prevention and Mitigation are the major activities in the pre-disaster stage. Prevention is always better than cure, so this program adheres to the principle. In this stage, various preventive measures and activities are undertaken well in advance so as to respond to disasters in an effective way.
- If the early warnings can be seriously heeded, much of the disaster could have been be avoided. The focus should therefore be on preparing communities and buildings to withstand disasters. There are events of cyclones in India quite frequently, and warnings are generally given before the event occurs. If operative activities can be undertaken well in advance, then it becomes easy to prevent huge loss in terms of lives and property, the aftermath phase.

During-disaster Stage:

- Response and Relief are the important activities in the during-disaster stage. It starts in the aftermath of a disaster.
- The immediate aspects of an aid response include the search, rescue, and evacuation, identification of dead bodies, the disposal of debris, providing first aid, food, water, shelter, safety and security, health care, and sanitation, and restoring basic facilities.

Post-disaster Stage:

- The major activities in the postdisaster phase include: Rehabilitation, Reconstruction and Recovery. Disasteraffected communities will be made resilient and able to return to normalcy through these activities**.**
- Generally, this phase takes a long time as the efforts are made to restore all essential facilities to pre-disaster status.
- The major focus of this phase is on measures that could pave the way for the long-term recovery of social, economic, and physical structures, as well as processes in such a way that future disasters are unable to impact severely and irreversibly.

Disaster preparedness programmes aim to improve the technical and management capabilities of governments, organisations, and communities in order to reach a suitable degree of readiness to respond to any emergency crisis. These precautions, which are described as being logistically prepared to handle disasters, can be strengthened by having reaction mechanisms and processes, practising them, creating long- and short-term strategies, educating the public, and constructing early warning systems.

To build a disaster-resistant house, you can consider the following techniques:

- **Reinforced structural design**: Use steel frame construction, reinforced concrete foundations, and proper bracing to improve the structural integrity of your home.
- **Flexible design**: Buildings with flexible designs can absorb seismic energy and deform without collapsing.
- **Lateral load-resisting systems**: Incorporate systems like shear walls or braced frames to distribute seismic forces and stabilize the structure.
- **Elevated structures**: Raise the building above potential flood levels to minimize water damage.
- **Flood-resistant materials**: Use materials that can withstand water damage, such as concrete or masonry.
- **Effective drainage systems**: Install efficient drainage systems to prevent water accumulation.

You can also consult a professional architect or engineer to create a robust design that accounts for potential natural hazards

How to Build a Home Resistant to Natural Disasters

Designing and building a home that will house special memories with family and friends for years to come, and adding a few extra steps to ensure it is protected is a worthy investment. A crucial part of building your dream home is considering the potential threats posed by natural disasters that could impact the structure and longevity of your home. Discover our key recommendations for *How to Build a Home Resistant to Natural Disasters*.

Location, location, location:

The most important factor to consider when creating a disaster-proof home is where you plan on building. Choosing the right site can significantly reduce the risk and impact of natural disasters. Research the area's history of natural disasters such as floods, hurricanes, earthquakes and wildfires.

Where possible, avoid building in flood zones (check out our blog on building in a flood zone), coastal areas prone to storm surges, earthquake-prone regions, or high-risk wildfire zones. Instead, opt for elevated sites that are less vulnerable to flooding or landslides.

Resilient building materials:

The choice of building materials plays a vital role in constructing a disaster-resistant home. The aesthetic look of your home is important but selecting durable materials that can withstand the forces of nature is more important.

Recommended materials include reinforced concrete, brick and steel. These all offer better structural integrity and resistance against wind, water and fire damage. Additionally, consider impact-resistant windows and doors, which can tolerate high winds and flying debris during hurricanes and tornadoes.

Reinforced structural design:

A well-designed and reinforced structure is key to building a disaster-resistant home. Work with a professional architect or engineer to create a robust design that accounts for potential natural hazards.

Implement techniques such as reinforced concrete foundations, steel frame construction and proper bracing to enhance the structural integrity of your house. This will ensure that your home can withstand the powerful forces exerted by earthquakes, hurricanes and high winds.

Waterproofing and drainage systems:

Water damage is a common consequence of most natural disasters. To mitigate this risk, incorporate effective waterproofing and drainage systems into your home's design.

Install a high-quality waterproofing membrane in the foundation and walls to prevent water intrusion during floods or heavy rainfall. Ensure proper grading and slope away from the house to direct water flow away from the foundation. Additionally, consider installing a sump pump system to remove excess water during flooding.

Wind and storm protection:

Hurricanes and high winds can cause significant damage to homes. When building a disaster-proof home, it is essential to incorporate wind and storm protection measures.

Use impact-resistant windows and doors that meet the necessary standards for hurricane zones. Reinforce roof connections to the walls with hurricane straps or clips. Choose a hip or gable roof design that offers better wind resistance compared to a flat roof. Secure outdoor structures such as decks, sheds and carports to prevent them from becoming flying debris.

Fire-resistant strategies:

Wildfires pose a severe threat to homes located in fire-prone areas. For a fire-resistant home, non-combustible materials are the safest choice for the exterior, such as metal or fiber cement siding.

Ensure proper venting and screening to prevent ember entry. Clear vegetation and maintain a defensible space around the house by removing dead plants, leaves and branches. Install fire-resistant windows and exterior doors. Consider using fire-resistant roofing materials such as metal or tile to reduce the risk of ignition.

Emergency preparedness:

Even if your home is well built and disaster resistant, it is important to be prepared for any emergency. Develop an emergency plan and educate your family on evacuation procedures, emergency contacts and the location of emergency supplies.

Install smoke detectors, carbon monoxide detectors and fire extinguishers in strategic locations throughout the house. Consider investing in a backup power generator to ensure electricity during power outages. Additionally, create a designated safe room or shelter within your home that can provide protection during severe weather events.

Insurance and documentation:

Obtaining comprehensive homeowners' insurance that covers natural disasters is crucial for protecting your investment. Review your policy and ensure it adequately covers the specific risks in your area. Keep a list of your belongings and important documents safe digitally, making it easier to file insurance claims in the event of damage or loss.

Regular maintenance and inspections:

Maintaining a disaster-resistant home requires regular inspections and maintenance. Schedule annual check-ups for your home's structural integrity, electrical systems, plumbing and roof. Address any identified issues promptly to prevent further damage.

Keep gutters and downspouts clear of debris to ensure proper water drainage. Trim trees and remove dead branches that could pose a risk during storms. Regular maintenance will help ensure your home remains resilient over time.

Here are some ways to reduce population vulnerability and increase awareness about disasters:

- **Identify vulnerable populations**

  Identify the communities and groups that are at a higher risk of poor health and develop strategies to reduce their vulnerability.

- **Educate the public**

  Use public awareness and education (PAPE) messages to help communities learn about disaster risks and how to stay safe.

- **Involve the community**

  Involve residents in the planning and implementation of disaster risk reduction initiatives.

- **Use media**

  Use memes, TV, and print advertising to spread awareness quickly.

- **Protect children**

  Schools play a critical role in promoting disaster prevention and empowering young people to understand and act on the risks they face.

- **Consider migration**

  Migration can expose people to new hazards, especially when it concentrates people in high-risk areas.

- **Consider urbanization**
  Urbanization can pose challenges to disaster response, especially for vulnerable groups like older adults.

Vulnerable populations are groups that are at a higher risk of poor physical, psychological, or social health. Factors that can increase vulnerability include:

- Social, economic, political, and environmental components
- Limitations due to illness or disability
- Low socioeconomic status
- Lack of social networks or family connections

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Lecture No 5

Rescue training for search and operation at the national and regional level can include:

- **Understanding hazards**: Learning about the risks and hazards in a structural collapse environment
- **Survey of affected areas**: Learning how to survey areas that have been affected
- **Basic life support**: Learning how to perform basic life support measures and provide pre-hospital care
- **Understanding rescue support systems**: Learning about rescue support systems at the regional, national, and international levels
- **Linking community response**: Learning how to connect community-based response to local emergency services

Search and rescue operations involve three separate steps:

- **Sizeup**: Assessing the situation and developing a safe action plan
- **Search**: Locating victims and documenting their location
- **Rescue**: Using procedures and methods to extricate the victims

Emergency response training can also be important for people working in offices, businesses, industries, and factories. This training can help people learn how to respond to emergency situations and can be the difference between life and death.

INTRODUCTION

Search, rescue and evacuation processes need to be carried out immediately after a disaster strikes a certain area or building. These are the most immediate critical operations that are usually performed by the local volunteers, voluntary organisations and the emergency agencies. Light Search and Rescue is a procedure carried out at primary stages, initially to find out persons with injuries in lightly damaged buildings, or even without any injuries and needing assistance, and to help them exit. If the condition worsens and the local groups are not able to control the situation, then the specialist groups within emergency agencies have to be called in for professional help, and at times even the Defence Forces including the Army, the Navy, the Air Force and the Coast Guard are called on for help. Primarily, Search and Rescue Operations are undertaken to save the maximum possible number of victims who are trapped in an area affected by a disaster. The basic aim of all such operations is to ensure the survival of the maximum possible number of affected people. A plan is worked out with the help of local people through surveys and then appropriate steps are taken by the various teams involved to carry out the operations. Besides physical rescue, the aim is also a systematic and organised approach in a post-disaster situation riddled with chaos and confusion.

SIGNIFICANCE OF SEARCH AND RESCUE (SAR)

Search and rescue, often known by the acronym SAR, is the process of identifying the location of disaster victims who may be trapped or isolated, and bringing them to safety and providing them with medical attention after a disaster strikes. Search and rescue generally involves the local people who are well versed with the local terrain and can be instrumental in searching

and accessing the trapped victims. SAR teams rely on sniffer dogs that are specially trained to smell out human beings trapped under the rubble. Heavy machines such as cranes and earthmovers are used to remove heavy rubble; and special equipment to delicately remove fallen structural elements and reach out inside heaps of rubble with visual or sound equipment for locating survivors. In case of floods and cyclones, boats and helicopters are used to carry out the search and rescue operations by forming teams and carrying out SAR operations in the entire area systematically, each team covering its assigned sectors. After the search, rescue and evacuation, some important steps are required to be taken in order to provide relief to the evacuees.

Prime amongst these are:

Medical Aid Water and food Shelter

Temporary subsistence supplies

Health and sanitation

Information,

and Security.

PHASES OF SAR

SAR is carried out for different categories of affected people depending on the complexities of the processes involved. This is illustrated as follows:

1) The immediate stage is searching and rescuing people with injuries among people trapped in the affected area that we see at first sight.

2) The second stage is the rescue of persons whom we are able to contact but are unable to reach. These are people who may be trapped under heaps of rubble from a collapsed building, or people stuck on trees or high-rise buildings, or trapped inside basements.

3) The third stage includes searching those areas where survivors are likely to be trapped generally. These are called likely survival points.

4) The fourth stage is the last stage in which debris leading to likely survival points is cleared.

LOGISTICS AND METHODS

Search and Rescue Kits

SAR kits should ideally be kept in central locations in areas of vulnerable buildings. Local trained personnel can make use of these kits should a disaster strike the area. A typical SAR kit will comprise the following items:

Evacuation map of the building or area

Hammer Screw driver (6" flat)

Axe 24"

Crow bar

Spade

Pickaxe

50-foot rope

Torch

Spare battery cells

Hard shoes or Gum Boots

Helmet

Hand gloves

Dust Mask

Simple Rescue Methods

Rescue can be carried out ideally by using rescue equipment but also by ordinary methods when equipment is not available. For rescuing an affected individual, there are two types of methods:

Single Person Methods and Two Person Methods.

A) Single Person Methods

Firemen's Lift

This method is used when the affected person is unconscious and only one rescuer is available for carrying out rescue work. In this method, the rescuer first makes the affected person lie on his or her the abdomen. Then he uses both his hands to lift the affected person by his armpits. He holds the right hand of the affected person with his left hand and with his free hand he holds the affected person's right leg. Now the rescuer lifts the affected person with both his hands and positions him or her in such a manner that his or her waist lies just above the neck of rescuer.

Human Crutch

This method is used when affected person has an injury in one of his/her leg. In this method, the rescuer first ties the injured leg of the affected person with his opposite leg with a handkerchief. If the affected person's left leg is hurt, the rescuer will tie this leg with his own right leg. The rescuer holds the affected person's waist with his hand and then they both can walk slowly with the rescuer supporting the affected person's injured leg with his own leg.

Pick a back

This method is used when the affected person has an injury in his or her leg and is not able to walk long distances, but can support himself or herself with his or her hand. In this method, the rescuer first gets down on one knee and allows the affected person to saddle on his back. The rescuer then holds the affected person at the thighs and the affected person encircles the rescuer's neck for support.

Staircase drag

This method is used when one has to rescue an unconscious person from a floor higher than the ground floor. In this method, the affected person is laid down on his back and his hands and feet are tied with a handkerchief. The rescuer holds the casualty's armpits and slowly drags him or her down the stairs head, first.

Firemen's Crawl

This method is used when the affected person is trapped inside a smoke filled area and is unconscious. The rescuer first ties the hands of the casualty and then goes astride the casualty. The rescuer inserts his head in the loop made by the casualty's hands. Then by pushing his hand and foot against the ground, he can drag the casualty to a safer place.

Tow Drag

This method is used when the affected person is lying in such an area where there is very little space to carry out the rescue. The affected person may or may not be unconscious. In this rescue method, the rescuer inserts his feet into the casualty's armpits and then pulls the casualty out of the confinement area. After taking the casualty out, the rescuer can lift the casualty with any appropriate method.

B) Two Person Methods

Fore and Aft

This method is used when the affected person is having an abdominal injury. For carrying out this method, two rescuers are required. In this method, the affected person is laid down flat on his/her back. One of the rescuers holds the casualty from the armpits while the second rescuer holds the legs. Then they both lift the casualty and in this way they can shift the casualty to a safer place.

Two Hand Seat

This method is used when the affected person is injured in one leg. For carrying out this method two rescuers face each other on either side of the affected person. They both bend and place their inside arms under the casualty's back just below the shoulders, raise him and put their outside arms under his thighs, holding each other's hands with a hook grip. The affected person can place his/her hands around rescuers' necks.

Three Hand Seat Method

This method is used when the rescuers need one hand free to support the affected person's injured leg. In this method, if the affected person's left leg is injured, the rescuer on the right grasps his own left wrist with his right hand & the other rescuer's left wrist with his left hand. The second rescuer will hold the right wrist of first rescuer keeping his left hand free to support the casualty's injured leg or any other medical equipment.

Four Hand Seat Method

This method is used for an affected person who is heavy and who can support her/ his self with his/ her hand. In this case, each of the two rescuers grasps his left wrist with his right hand, grasping the other rescuer's right wrist with his left hand.

Carry Chair

This method is useful to lift a handicapped or a very old person. The person to be lifted is placed on a chair, and then, the chair can be carried to a safer area.

Knots and Lines

A rope is an important part of SAR operations. It can act as a guideline during searching; it can be used to lift equipment during rescue work or while carrying out fire fighting, and it can be used for rescuing people. Various knots and lines are used, which are mentioned below:–

Over hand knot or thumb knot

It is used as a stopper knot. Sometimes it is also known as first aid knot because it is used to tie the bandage.–

Figure of Eight

It is used to stop a line running through a sheave or a pulley.–

Reep Knot

It is used to join two ropes of the same diameter or thickness.–

Chair Knot

It is used to lower an unconscious casualty.–

Half Hitch

It is used to hoist round object.–

Claw Hitch

It is used to secure a line to a round object.–

Timber Hitch

It is a simple knot with several loops on it, which is used to hoist any odd heavy object.–

Black Wall Hitch

It is used to secure a line to the shank of a hook.

– Draw Hitch

It is used as temporary fastening when we require the knot to be released immediately.

– Single Sheet Bend

It is used to join two ropes of different thickness.–

Double Sheet Bend

It is similar to single sheet bend but with a double hitch round.–

Carrick Bend

It is used to join two lines without the knots forming an obstruction to pass round an object.–

Sheep Shank

It is used to shorten the length of a line or to avoid any damaged part of the line.–

Single Bowline

It is used for various miscellaneous purposes.–

Round Turn and Two Half Hitch

It is used to secure a line to round objects.–

Cats Paw

It can be used to hoist a hose or a similar object.–

Running Bowline

It consists of a bowline, with the standing line passing through the bight of the line, to form a running noose.–

Bowline on Bight

It is used to lower a stretcher or an unconscious person.

Rescue Using Chair Knot The following safety precautions need to be taken while lowering a casualty with a line:

Always use a guideline to prevent the casualty from striking with building wall; The line should pass hand to hand, and not through hands;

The line should run over a wooden block or some sack under the point of contact of line and building;

Check the area for required space;

Insert the shorter loop into casualty's armpits and longer loop into the casualty's knees;

While lowering the casualty assign someone on watch below the casualty.

Search and Rescue Checklist

Finally, the following checklist, as per the Source book on District Disaster Management, Ministry of Agriculture, Government of India, and National Academy of Administration, Mussourie can help assess the level of preparedness of search and rescue operation plans. What rescue tasks may need to be performed? Who is responsible, who coordinates? Are there procedures for detecting and marking danger areas? How are search and rescue activities integratedwith other emergency functions, particularly in health?

BEHAVIOURAL REQUIREMENTS

Qualities of a Good Rescuer A good rescuer needs to be physically fit and trained/experienced to be able to effectively carry out rescue work. Incapable rescuers usually hinder the work on site, and often also cause more harm by aggravating the situation and causing further injuries to the persons to be rescued. The following key features are seen when selecting rescuers for a mission:

Technical ability

Common sense

Directional sense

Effective improvisation

Proper judgment Physical and mental fitness

General Precautions while carrying out Search and Rescue

SAR is a process with high level of risk to the rescuers as well. It is important for the rescuers to take adequate precautions to keep themselves safe. The following actions are advised:

Wearing necessary personal protective equipment

Always working in pairs Before starting SAR, one should always note down alternative means of escape for oneself Using guidelines such as directions for rope use when you enter a confined place, smoke filled room or a dark place

Carrying out thorough searches in toilets, storerooms, under tables, in lifts, etc.

Evacuation

Evacuation implies removing all people from a threatened area to a safer place, before, during or after an emergency. It has been observed that in most disaster events, like earthquakes, cyclones and fires, a large percentage of total deaths occur due to wrong evacuation practices or stampede. It is necessary to evacuate using short and safe exits. Generally, those areas should be avoided that make occupants more vulnerable.

Evacuation in Slow and Rapid Onset

Disasters Mass evacuations are carried out to move people out of impact areas in case of cyclones or floods, once the warnings have been received. People from expected areas of impact are moved to cyclone shelters or schools or other public buildings in nearby places that are designated, emergency shelters. Public vehicles are often organised to make such evacuation possible. The general public is warned of the coming disaster, and advised to evacuate themselves or take the help of the government or NGOs working in the area to move out. Evacuation is more difficult in very rapid onset disasters such as earthquakes, fires or accidents. Mass evacuation is not possible in such cases since there is very little or no warning time available. In such cases people caught in the event can just follow preplanned evacuation systems and use exit signs, exit routes, meeting points etc. Preplanning for evacuation is most critical for such situations. Such planning is discussed in the following sections:

Evacuation Planning

An Evacuation Plan is a plan that shows the shortest and safest exit routes and the location of first aid, fire fighting equipment and SAR equipment in a building or area. The evacuation plan should be prepared in non-disaster times, and should be kept updated. It should also be used in mock drills. To check the level of preparedness provided for in evacuation planning, Evacuation Checklist has been provided in the Source Book on Disaster Management, Ministry of Agriculture, and Lal Bahadur Shastri National Academy of Administration, Mussourie.

Evacuation Checklist

Does any person or organisation have the authority to evacuate people?

Are there designated locations to which evacuees should travel?

How many people may need to be evacuated?

In what circumstances should they be evacuated?

Who will tell people, that it will be safe to return?

Who will trigger this?

Are staging areas and pick -up points identified for evacuation?

Are evacuees to be provided with information or where they are going and how they will be cared for?

Is there security for evacuated areas?

How are prisoners to be evacuated?

How are the cultural and religious requirements of evacuees to be catered for?

Who is responsible for traffic control during evacuation?

How are evacuees to be registered?

The process of making an evacuation plan is as follows:

First take the building or area plan as Base Map Make separate floor plans

Remove the unnecessary text or dimension from the map

Show the Exit Route by arrow signs from each room to the assembly point or points in easily accessible open spaces.

Try to provide maximum number of exits.

Put 'Emergency Exit' labels at the functional exits

Show the locations of fire extinguishers

Locate and highlight the site for SAR and First-Aid Equipment Mark safe open areas as assembly points

Locate 'Danger Areas' and mark them with red signs with skeleton symbols

Separately draw room exit plans and display them in individual rooms

There should be a provision for at least two exits for each room.

Locate room exits in opposite direction to the extent possible.

Importance of Exit Signs

Exit signs help in getting out of a building during an emergency. As we know, during emergencies people panic and some times forget the exit routes. During such times, exit signs help people get oriented and exit the danger areas. Getting out is not enough in itself; getting out safely is important. Often people get injured and stampedes are caused while getting out. Prominently visible exit signs help people out even when it is dark.

Pre-Planning for Evacuation

An order to evacuate could come at any moment. By planning ahead, individuals can evacuate quickly and safely without sacrificing life, limb and property. The goal should be to spend as little time as possible following an evacuation order.

Evacuation Activities

Lock door and windows and turn off utilities like water taps, cooking gas and electricity.

Shut off systems that draw outside air, such as fireplaces and air conditioners.

If you can provide transportation for a neighbor who has none, do so.

Try to help the aged, disabled and other people around you who may need special attention, such as women and children.

In case of fire or earthquake, evacuate immediately.

In case of cyclone or any other evacuation where there is a little time, carry essential items, such as:

Medical supplies (prescriptions, first aid)

Disaster Management Money (cash, credit cards, important documents)

Personal hygiene items (washing, shaving, dental, eye-care, sanitary)

Clothing, bedding Baby needs (food, diapers, favorite toys)

Portable radio and batteries

Miscellaneous useful items (matches, flashlights, plastic bags)

Keep your radio tuned to a local station for emergency news updates.

Lecture No 6

Immediate relief in disaster management is the emergency act that takes place during and immediately after a disaster. It includes:

- **Warning and evacuation**
- **Search and rescue**
- **Providing immediate assistance**
- **Assessing damage**
- **Debris clearance**
- **Emergency accommodation**

Other aspects of disaster management include:

- **Preparedness**

  Initiatives to save lives, reduce property damage, and mitigate the impact of a crisis

- **Recovery**

  The process of returning the situation to how it was before the crisis, including clean up, identifying support, and long-term monitoring

- **Crisis counseling**

  Programs to help disaster-impacted individuals and communities recover through outreach and psycho-educational services

- **Financial assistance**

  Emergency relief organizations provide immediate financial support to people in financial crisis

- **Disaster risk reduction**
  Includes prevention, mitigation, and preparedness

Assessment

The purpose of assessments in the initial phase of a disaster is to provide information that can guide our emergency services in activities such as search and rescue missions, pinpoint location and nature of secondary threats, provide information about the status offacilities needed to treat or support the survivors

> The first step after stabilizing the situation by providing sufficient relief is to assess the damage. A meticulously executed assessment exercise would provide an ideal base for the rehabilitation efforts. This exercise is best carried out through multi-disciplinary teams which go into all aspectsof damage (social, economical, psychological) in participation with the local community. Based on the assessment of the damage and the needs, a recovery strategy has to be

formulated. The strategy should include all interventions - economic, social, political and psychological. The resources should be identified and the roles and responsibilities of all concerned should be defined.

Assessment surveys are used in disaster management to gather information and feedback to help guide emergency services and tailor relief efforts:

- **Needs assessment**

  Surveys can help identify the immediate needs of affected populations, such as food, shelter, and medical care.

- **Risk assessment**

  Surveys can help gauge the level of risk perception, preparedness, and understanding of disaster protocols within communities.

- **Post-disaster evaluation**

  Surveys can help assess the effectiveness of response efforts by gathering feedback from affected individuals.

- **Community engagement**
  Surveys can help involve communities in the planning process by seeking their input.

Post Disaster Rehabilitation phase starts immediately post the response stage, when order begins to be restored and an estimate of losses/ damages due to the disaster could roughly be assessed. Post Disaster Needs assessment is a prerequisite for planning disaster Rehabilitation.

Post-disaster rehabilitation is the process of restoring and improving public and community services in areas affected by a disaster. The goal of rehabilitation is to return community and government life to normal.

Rehabilitation can include:

- **Physical rehabilitation**

  Reconstructing physical infrastructure, such as land use planning, retrofitting, and watershed management.

- **Social rehabilitation**

  Restoring social structures, especially for vulnerable groups like children, women, the elderly, and marginalized communities.

- **Economic rehabilitation**
  Restoring livelihoods and business continuity, with an emphasis on sustainable employment and income generation.

Rehabilitation is part of the post-disaster recovery process, which also includes reconstruction. The recovery phase typically begins after the relief phase and can last for several years.

Some other activities that can be part of post-disaster rehabilitation include: Education, Health, and Habitat planning and improvement

Disaster Relief and Rehabilitation

Disaster Relief and Rehabilitation forms important components of Disaster Management. The phase of disaster relief provides essential requirements for survival of the affected community. Relief must be provided in a humane manner recognizing the dignity of individuals and their right to post disaster relief. Disaster Rehabilitation is part of Post-Disaster Recovery and is a long-term strategy that must help in the mainstreaming of Disaster Risk Reduction efforts into developmental planning .

Disaster Relief is the process of responding to a disaster event that has led to adverse impact on the lives of the affected community, to alleviate the suffering and providing humanitarian aid. Relief refers to the provision of appropriate essential services to the affected community in a timely manner based on the initial rapid assessment of the immediate requirements after disasters. The evacuation of people from their residence in cases of risk of the occurrence of disasters, and thereby provisions of emergency services including temporary shelter and immediate needs like food, clothing, non-food items etc. are also considered under disaster relief. In the alleviation of sufferings of the affected community, the phase of disaster relief is usually considered up to six months from the occurrence of the incident. It includes the immediate aftermath when essentials are required and extends up to the rehabilitation phase of disaster response/ recovery sometimes overlapping with it. Disaster Recovery needs to be synchronized with rehabilitation and reconstruction efforts in order to bring about quicker and effective recovery of the affected community. The following figure (Figure-1) indicates the Disaster Recovery cycle with an indicative timeline of the steps post disaster event and corresponding unit cost of materials. The First facet of disaster recovery is the Relief that starts with the disaster event and extends till stability is reached (usually about 6 months post the devastating event). This is followed by the Rehabilitation and Reconstruction phases that start post some stability is reached. Proper needs assessment is an important component of these phases and thereby the requirement for stability.

Disaster Rehabilitation

Rehabilitation phase starts after the completion or near completion of immediate response stage, when order begins to be restored and an estimate of losses/ damages due to the disaster could roughly be assessed.Rehabilitation may be defined as “an overall dynamic and intermediate strategy of institutional reform and reinforcement, reconstruction and improvement of infrastructure and services, aimed towards the support of the initiatives and actions of the affected population in the political, economic and social domains, as well as reiteration of sustainable development.” (NDMA, 2016). The efforts on post disaster reconstruction and rehabilitation (that are important facets of disaster recovery) are considered as opportunities to “build back better”. They are seen as enablers for mainstreaming disaster risk reduction efforts and embedding Risk reduction into developmental measures for building resilient communities.

Rehabilitation can be classified into four categories:

1. Physical Rehabilitation– Reconstruction of physical infrastructure which includes land use planning, micro and macro-zonation, retrofitting, Disaster resilient cropping patterns, watershed management etc. It also includes the aspects of **Relocation**.

2. Social Rehabilitation– These includes restoration of social structures in the destruction areas and may emphasize upon the vulnerable groups like children, elderly, women and marginalized communities.

3. Economic Rehabilitation – includes livelihood restoration and business continuity, so that income generation is restored with particular emphasis on sustainable employment/ income generation.

4. Psychological Rehabilitation– includes stress management with particular focus on Post-Traumatic Stress Disorders

Rehabilitation programs must include the understanding developed over time about the systematic causes and the long-term consequences of the disaster event (Niazi). The vulnerability of the community that was exacerbated by the exposure to the disaster event must be addressed in this stage of disaster management by mainly addressing the altered poverty levels emerging after the disaster. Immediate relief and reconstruction activities including physical rehabilitation, provides immediate boost to the local economy. When such physical measures are supported by the improvement in land, water and environmental resource management, with improvement of economic opportunities, the vulnerabilities could be addressed effectively, thus ultimately improving the quality of life and disasters thus provide the window of opportunity during the rehabilitation phase to gradually introduce to the community concepts like better construction practice, rainwater harvesting, improved sanitation, sewerage and drainage thereby promoting better management of resources. Figure-3 indicates the steps from Disasters to sustainable development.

Rehabilitation must, therefore, be undertaken taking into consideration the detailed socio-economic profile of the affected community and risk assessments with monitoring and evaluation frameworks in place. Past interventions (pre-disaster development related schemes and programs and post-disaster interventions) must be monitored and analyzed to avoid creating or exacerbating risks. A community led process, in coordination with local staff would ensure proper planning and implementation of rehabilitation programs. The local government thus plays a very important role in the rehabilitation of disaster-affected community.

PDNA & Rehabilitation

Post Disaster Needs Assessment is a prerequisite for planning disaster Rehabilitation (IFRC, 2012). A detailed needs assessment would bring about the requirement for interventions at various levels. Based on these identified requirements, rehabilitation strategy is prepared that considers the following :

The nature of disaster and the resultant type, scale and extent of damages Proper stakeholder analysis

Coordination between various agencies involved in rehabilitation

Risk management to ensure new risks are not built into the community during the rehabilitation/ development phase and focusing on Disaster Risk Reduction efforts Gender sensitive redevelopment

Environmental and other resource considerate redevelopment in the form of interventions that bring about sustainable improvement in the quality of life.

Balancing needs with available resources are critical in the rehabilitation stage. Relief stage might attract large international and national aid while the rehabilitation stage might not garner the same amount of attention. Thus prioritization of investment would be a key requirement for effective rehabilitation. Bottlenecks for rehabilitation include (Aysan & Davis, 1993):

Funds– sources and distribution

Materials for reconstruction, livelihood diversification or resource management

Equipment and tools– for facilitation of reconstruction and rehabilitation

Energy/ Power– for critical infrastructure and facilitate economic rebuilding of affected community

Land– especially post devastating disasters like landslides, floods, soil/ sea erosion etc.

Human resources– due to casualty or outmigration

Data– reliable information could be problematic due to the decentralized work environment and lack of inter agency coordination

Administrative structures and organizations– due to excess work pressure on public sector

Lecture No 7

The rehabilitation stage of post-disaster recovery is when basic services and facilities are restored for a community or society affected by a disaster. The rehabilitation stage is typically part of the recovery phase, which can last for several years.

Here are some political and administrative aspects of the rehabilitation stage:

- **Community-government partnership**

  Past experiences show that rehabilitation efforts are only sustainable with a partnership between the community and the government.

- **Role of NGOs and international organizations**

  The role of NGOs and international organizations is reduced after a certain period. However, they can provide expertise in organizing rescue and relief operations.

- **Training**

  A training of trainers center can be developed at the district level to train NGOs in long-term rehabilitation efforts.

- **Building codes**

  Building codes can be updated every five years and made mandatory in all state and municipal building bye-laws.

- **Monitoring and compliance**

  A sound compliance regime with binding consequences can be put in place to ensure the effectiveness of techno-legal and techno-financial provisions.

Other aspects of the rehabilitation stage include:

- Physical rehabilitation, such as reconstruction of physical infrastructure
- Social rehabilitation, such as restoration of social structures in the destruction areas
- Economic rehabilitation, such as livelihood restoration and business continuity
-

Social Rehabilitation

In the post-disaster phase, family support systems can break down due to physical and mental trauma resulting from losses of life and property, physical dislocation, and migration of some members of disaster affected communities.

Social aspects of post-disaster rehabilitation include:

- **Providing support**

  Vulnerable groups like the elderly, orphans, and single parents with young children need special social support to survive.

- **Building infrastructure**

Building community centers, day care centers, old age homes, and other infrastructure is important.

- **Providing aid**

  Providing shelter, food, financial assistance, and other forms of aid is important.

- **Addressing psychological issues**

  Strategies are needed to help people overcome psychological and psychosocial problems.

- **Strengthening social capital**

  Placemaking can help strengthen social capital and address uncertainty about the future.

- **Training social workers**

  Trained social workers can use participatory approaches to help vulnerable people and reinforce social capital.

- **Protecting vulnerable groups**

  Social workers can help protect vulnerable groups and prevent their needs from being neglected.

- **Providing assistance to farmers**
  Farmers may need assistance with sowing and harvesting, and may need cash grants or loans to purchase seeds and fertilizers.

Other aspects of post-disaster recovery include:

- Health and sanitation
- Livelihoods, food security, and economic recovery
- Community infrastructure and services
- Disaster risk reduction
- Local capacity development

Lecture No 8

The economic aspect of post-disaster rehabilitation involves restoring livelihoods, introducing new economic opportunities, and improving land and water management. Here are some things to consider:

- **Economic impact study**

  Assess the physical and economic damage, including job loss, business closures, and tax revenue loss.

- **Business recovery centers**

  Set up centers to centralize resources for maintaining business continuity.

- **Training**

  Provide training to make women economically independent, such as tailoring, computer handling, and self-help groups.

- **Residential female children homes**

  Provide education and training to make young girls self-reliant and economically independent.

- **Community centers**

  Provide meeting places, counseling, and training for women.

- **Creches**

  Provide creches for the children of women who are working or engaged in social activities.

The post-disaster recovery process is dynamic and non-linear, and can be broken into two periods: short-term and long-term. The short-term phase typically lasts from six months to at least one year, while the long-term phase can range up to decades

The environmental aspects of post-disaster rehabilitation include:

- **Rapid Environmental Assessment (REA)**

  Evaluating the damage to the environment, including the impact on air, water, land, and other natural resources

- **Waste management**

  Clearing, sorting, recycling, and disposing of disaster debris in an environmentally friendly way

- **Water contamination assessment**

Evaluating the contamination and pollution of water sources, such as wells, rivers, and ponds

- **Hazardous materials handling**
  Managing hazardous and toxic materials, such as oils, paints, petrol, and chemicals, to avoid polluting the air, water, and land

Other aspects of post-disaster rehabilitation include:

- **Physical rehabilitation**

  Reconstructing physical infrastructure, such as houses, buildings, roads, railways, water supply, and electricity

- **Social rehabilitation**

  Restoring social structures in the affected areas, especially for vulnerable groups like women, children, the elderly, and marginalized communities

- **Economic rehabilitation**

  Restoring livelihoods and business continuity, with an emphasis on sustainable employment and income generation

- **Psychosocial recovery**
  Identifying concerns, setting priorities, and helping survivors make decisions